1　THU 2025

ジエステル A ：　$C_{10}H_{18}O_6$　$U = 8 = -\overset{O}{\overset{\|}{C}}-O- \times 3 + \bigcirc \times 1 + 1$

$KMnO_4$ で +30 は $-CH_3 \longrightarrow -COOH$

A $\xrightarrow{\text{けん化}}$ $\xrightarrow{H^+}$ $HO-(CH_2)_2-OH$

(芳) C (136) ： $C_8H_8O_2$　両1

C ： $H_3C-\bigcirc-\overset{O}{\overset{\|}{C}}-OH$　両2

$\xrightarrow[+30]{KMnO_4}$ パラ体 F ： $HO-\overset{O}{\overset{\|}{C}}-\bigcirc(Br)-\overset{O}{\overset{\|}{C}}-OH$　両2

D ： $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ (アセトン)　両3　$\xleftarrow{\Delta}$ $(CH_3COO)_2Ca$

ジカルボン酸 E

$E = A + 3H_2O - (C_2H_6O_2 + C + D) = C_3H_4O_4 = HO-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-OH$　両4

C ： $H_3C-\bigcirc-\overset{O}{\overset{\|}{C}}-OH$　$HO-CH_2-CH_2-OH$

E ： $HO-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-OH$

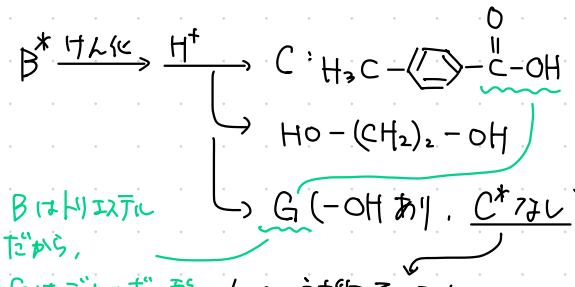
$CH_3-\underset{OH}{C}=CH_2$

D ： ↑ 分子量

$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

A ： $H_3C-\bigcirc-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-O-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-O-\overset{CH_3}{\overset{|}{C}}=CH_2$　両5

トリエステル B ： $C_{16}H_{18}O_6$　$U = 8 = -\overset{O}{\overset{\|}{C}}-O- \times 3 + \bigcirc \times 1 + 1$



B はトリエステルだから，G はジカルボン酸

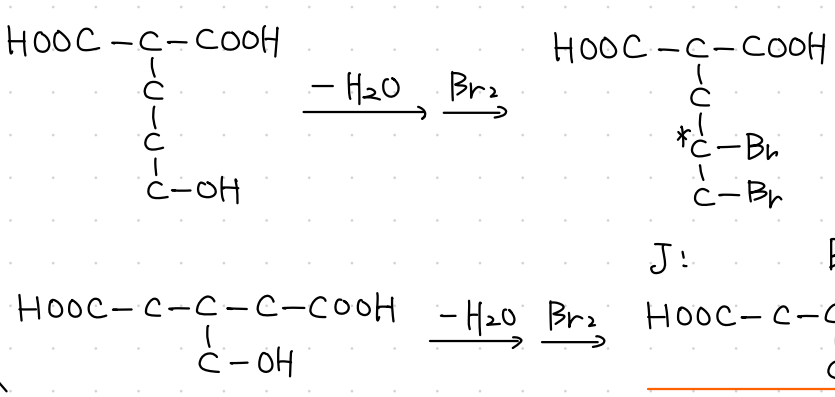
- 対称面あり
- $G = B^* + 3H_2O - (C + C_2H_6O_2) = C_6H_{10}O_5$　$(U = 2)$
- G (1級 -OH あり) $\xrightarrow[+14]{K_2Cr_2O_7}$ H (カルボン酸)
- G $\xrightarrow{-H_2O}$ I (C=C あり) $\xrightarrow{Br_2}$ J (C* なし)

$K_2Cr_2O_7$ で +14 は

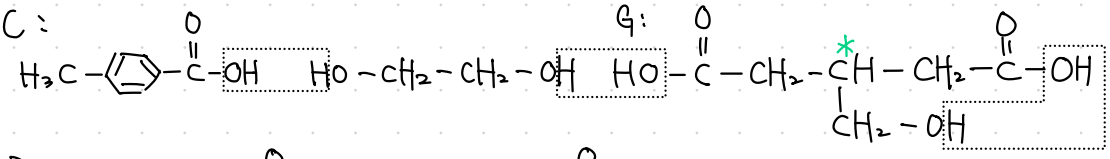
$-CH_2OH \longrightarrow -COOH$

1級アルコール

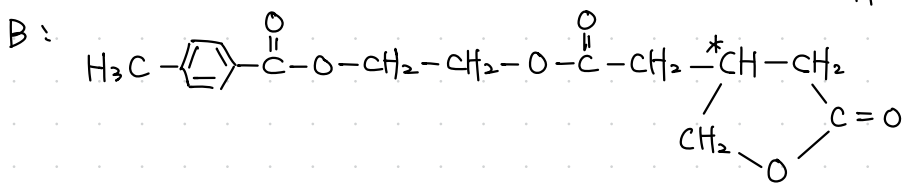
∴ G :



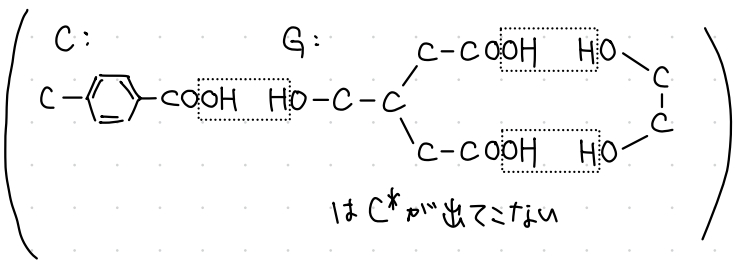
問6



環 ×1

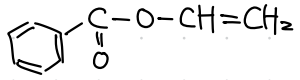


問7



問8　b. d　　問9　略

問10



問11　2.5 L　$\left(\frac{222\ g}{148n\ g/mol} \times n = 0.60\ mol/L \times V\ L\right)$

問12　(1) N : ビニロン ($\xleftarrow{けん化}$ M : PVA)

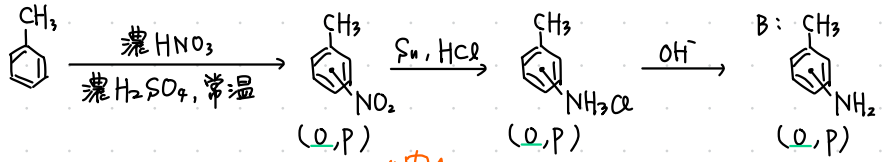
(2)　$\frac{22\ g}{44n\ g/mol} \times (44 + 6 \times 0.32)n \fallingdotseq \underline{23\ g}$

2　THU 2024

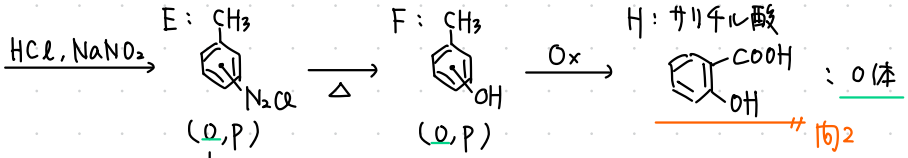
A^{**} : $C_xH_yO_zN_w$, C^* 2つは4級C原子ではない.

$A^{**} \xrightarrow[HCl]{\text{加水分解}} \xrightarrow{\text{中性}}$ ㊤ B, C^*, D (≦200)

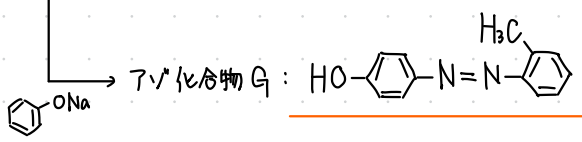
HCl →
- ㊤ C^*, D $\xrightarrow{NaHCO_3}$ ㊤ 中性 C^* / ㊦ カルボン酸 D の Na塩
- ㊦ アミン B の HCl塩



向1

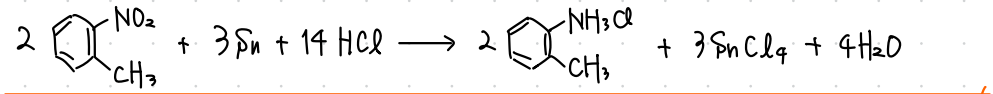


向2



向4

向3



1級アルコール

C* : $C_{10}H_{20}O$ # 問5 (U=1 : C=C×1) $\xrightarrow{-H_2O}$ J ($C_{10}H_{18}$, C*なし, 末端C=C×2) # 問5

↓ H_2

K (C*なし = 対称面あり)

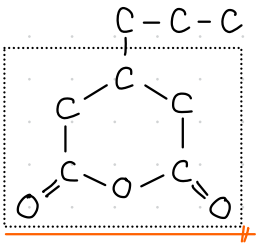
J $\xrightarrow{KMnO_4}$ D ($C_8H_{14}O_4$) + $2CO_2$ # 問7 (ジカルボン酸)

D $\xrightarrow{-H_2O}$ L (六員環・$\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}$ あり)

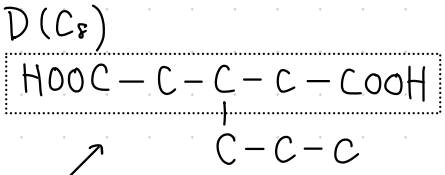
D + $HO-(CH_2)_2-OH$ → 高分子M

$-CH=CH_2 \xrightarrow{KMnO_4} -COOH + CO_2$

L : (C_8)

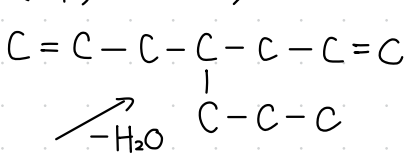


問8

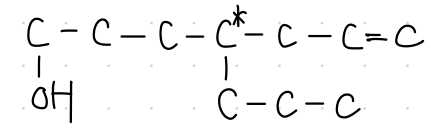


D (C_8)

J (C_{10})

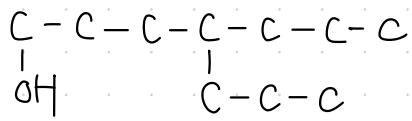


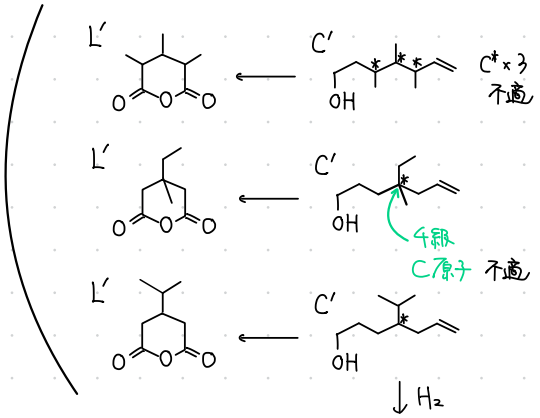
C (C_{10})



問6

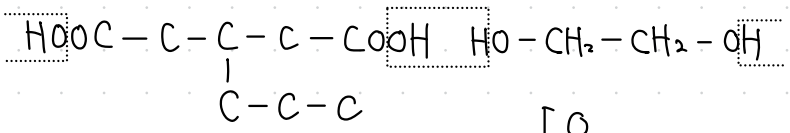
K (C*なし)

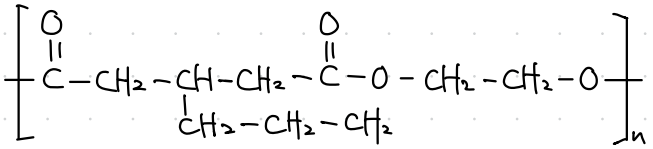




C*が出てきてしまう 不適

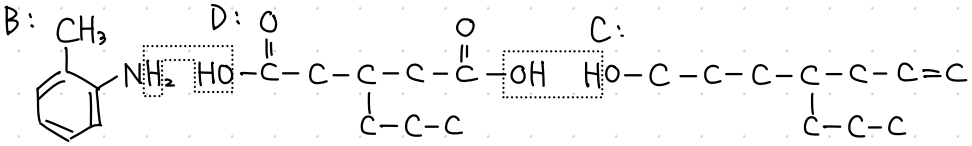
D :

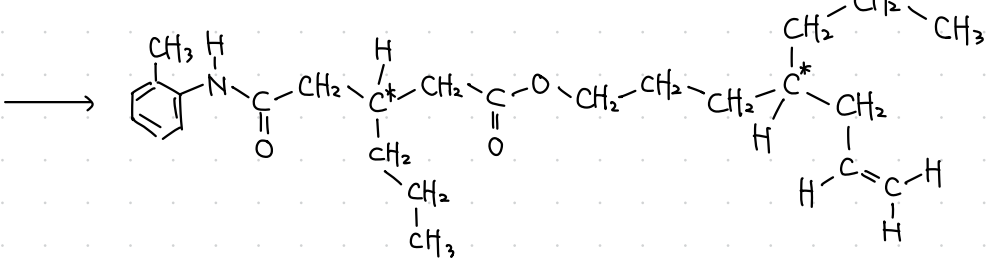




問9

B :

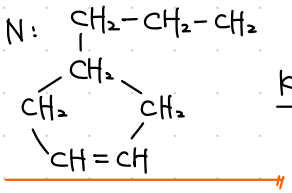




問12

例10　2.4×10^{2} 元

例11　C_8H_{14}　$U=2$

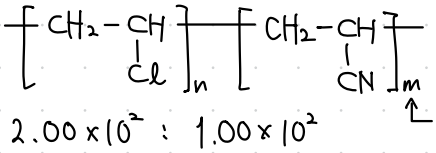
N： 

$\xrightarrow{KMnO_4}$

D：

$$HOOC-C-\underset{\displaystyle C-C-C}{\overset{\displaystyle |}{C}}-C-COOH$$

3 THU 2023

問4 (1)  (2) 縮合

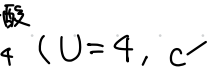
(3) $2.00\times10^{2} : 1.00\times10^{2}$

A $\xrightarrow{\text{けん化}}$ $\xrightarrow{H^+}$ B, C, D

B : CH_3-CH_2-OH $\xrightarrow[130\sim140℃]{-H_2O}$ E : $CH_3-CH_2-O-CH_2-CH_3$ #問1

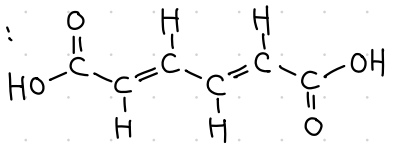
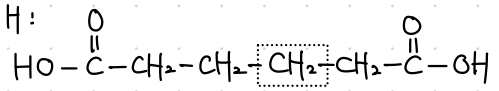
$\xrightarrow{K_2Cr_2O_7}$ F : $CH_3-\overset{\overset{O}{\|}}{C}-H$ $\xleftarrow{H_2O}$ G : $CH\equiv CH$

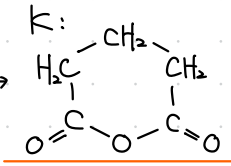
($CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + CH\equiv CH$) #問3

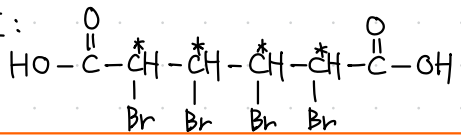
C : $C_6H_6O_4$ (シスムコン酸) ($U=4$,  , C=Cは全てトランス型)

$\xrightarrow{H_2\times2}$ H $\xrightarrow{-CH_2-\text{脱離}}$ J $\xrightarrow[\triangle]{-H_2O}$ 酸無水物 六員環K

$\xrightarrow{Br_2\times2}$ I

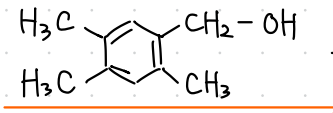
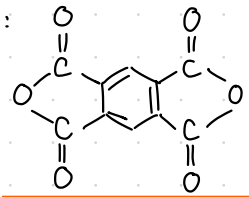
∴ C :  $\xrightarrow{H_2}$ H : 

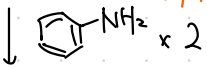
H $\xrightarrow{\text{脱離}}$ J $\xrightarrow{-H_2O}$ K :  #問6

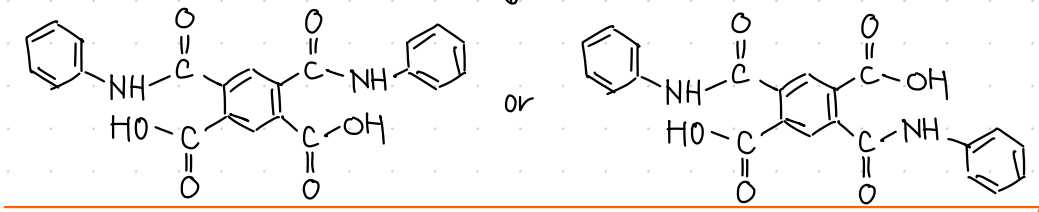
∴ I :  #問5

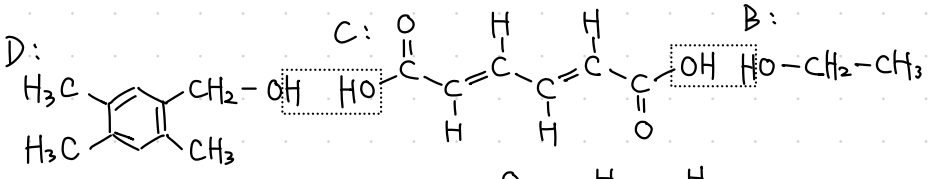
第1級アルコール

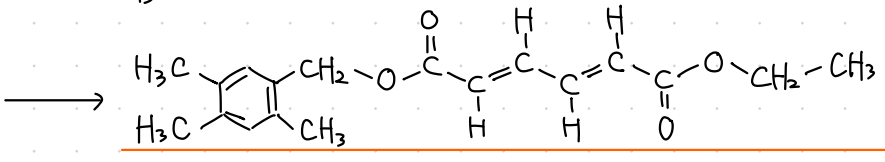
D :  4置換体 ($C_{10}H_{14}O$, $U=4=$  $\times1$) #問7 $\xrightarrow{Ox}$ L $\xrightarrow[\triangle]{-2H_2O}$ M

∴ D :  $\xrightarrow[\triangle]{Ox\ -2H_2O}$ M :  #問8 #問9



 #問10

D : 

 #問11

4

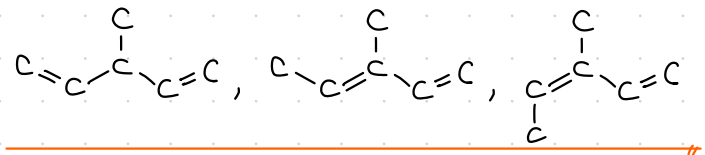
A $\xrightarrow{\text{けん化}}$ $\xrightarrow{H^+}$ ジカルボン酸 B 以上 ; $\underbrace{C_6H_{12}O}_{C,\ D}$ ($U=1$) 問1

アルコール C** { $CH_3-\overset{*}{C}H-$ あり

$\xrightarrow{H_2}$ E

$\xrightarrow{-H_2O}$ ～～

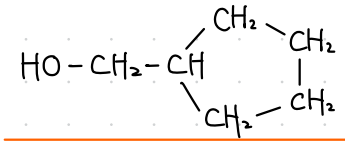
C : $CH_3-\overset{*}{C}H-\overset{*}{C}H-CH=CH_2$ (OH, CH_3) 問3

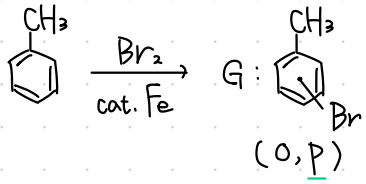
$\xrightarrow{-H_2O}$  問4

アルコール D { CH_3-CH- なし (OH)

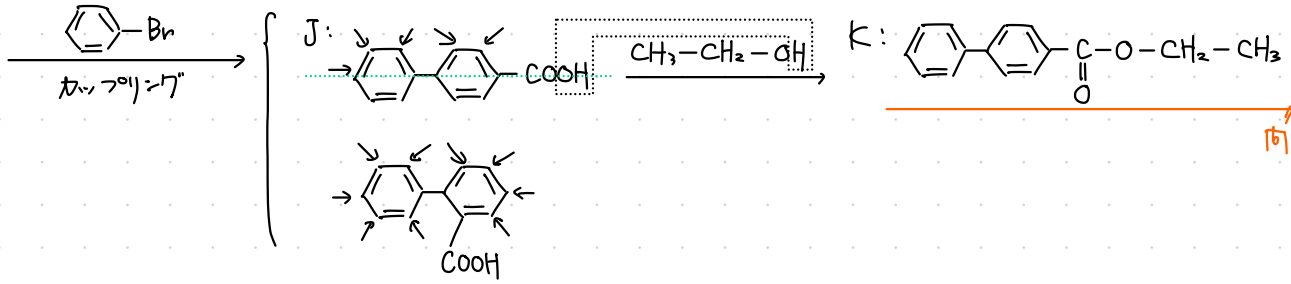
C原子5コ以上の5員環以上

$\xrightarrow{Ox}$ アルデヒド F

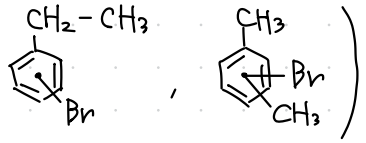
D :  問5

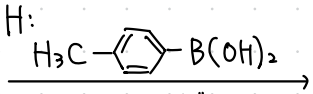


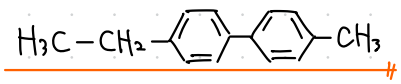
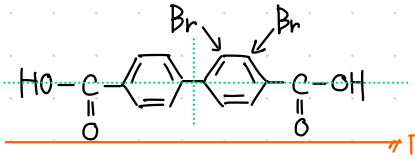
問7



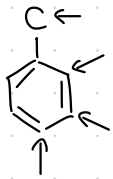
問8

L (C_8H_{10}, $U=4=$ ⬡ ×1) $\xrightarrow[\text{cat. Fe}]{Br_2}$ M ( ,)

H :  $\xrightarrow[\text{カップリング}]{}$ N $\xrightarrow{KMnO_4}$ ジカルボン酸 (以上) B $\xrightarrow{\text{⬡ の H を Br 置換}}$ iso × 2

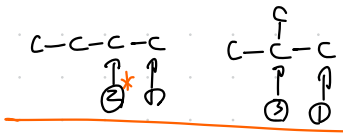
∴ N :  問9 $\longrightarrow$ B :  問10

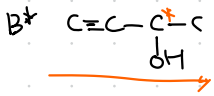
問2 略答

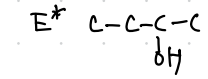
問6  4

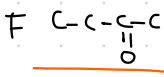
5

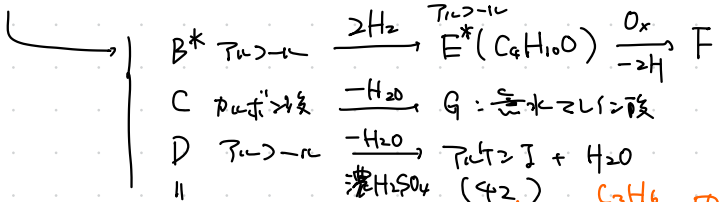
A : $C_{11}H_{14}O_4$ (U = 5)

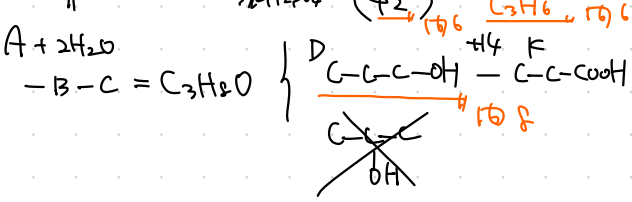


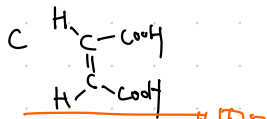


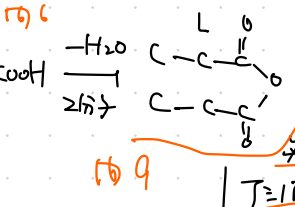


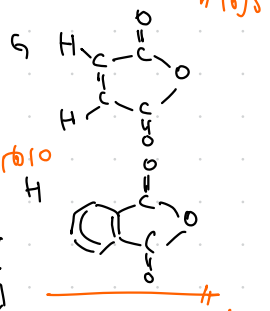


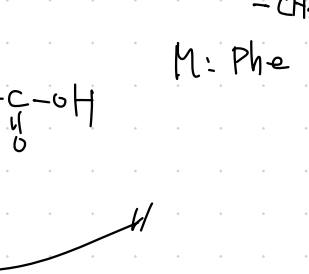


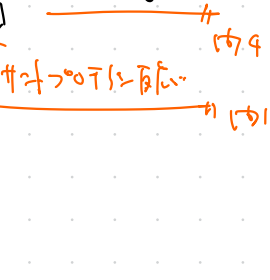










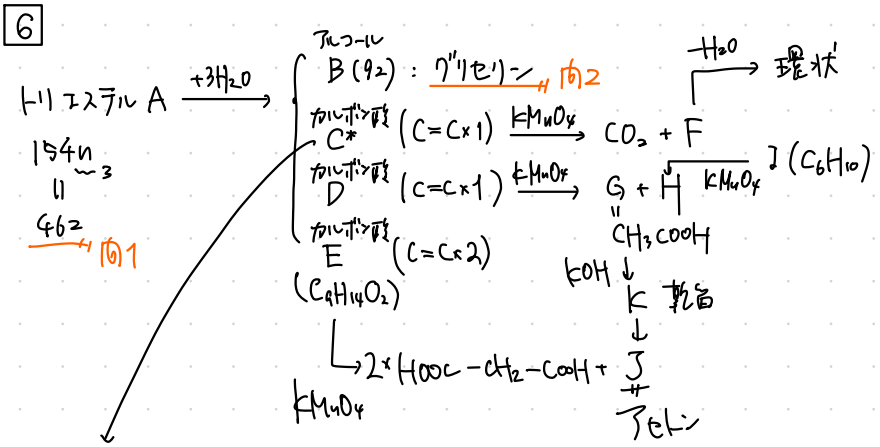


PP

M : Phe

キサントプロテイン反応

6



I: 

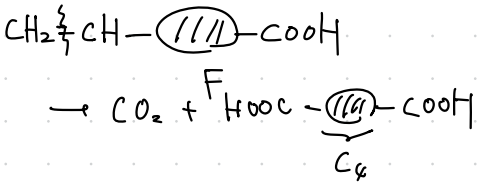
ア：マルトース
イ：グルコース
ウ：エタノール
エ：アセトアルデヒド
オ：酢酸 = G

H
$HOOC-(CH_2)_4-COOH$

D $CH_3-CH=CH-(CH_2)_4-COOH$

E $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}=CH-CH_2-CH=CH-CH_2-COOH$

C*: $C_nH_{12}O_2$ $U=2$



F: $HOOC-C-\underset{\displaystyle C}{\underset{|}{C}}-C-COOH$

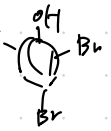
C: $CH_2=CH-C-\overset{*}{\underset{\displaystyle C}{\underset{|}{C}}}-C-COOH$

7

ケトン
A : $C_{13}H_{10}O$　$U = 9 = ⬡ \times 2 + -\overset{\overset{O}{\|}}{C}- \times 1$

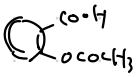
└→ エステル B のみ（Aは対称）

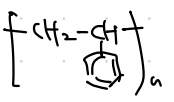
バイヤー・ビリガー酸化 → $+H_2O$ → 安息香酸 C ＋ 安息香酸 D

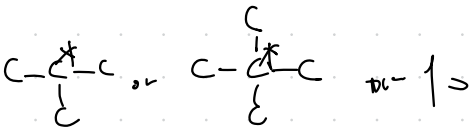
Br_2水 → 置換 → 

C：−COOHあり −OHあり

D：−COOHあり

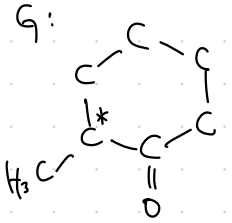
C → $CoNa_2$塩 （Na）→ 高温・高圧 CO_2 → ？ → $(CH_3CO)_2O$ → 

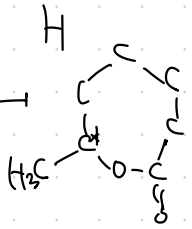
E :  → （付加重合） F : 

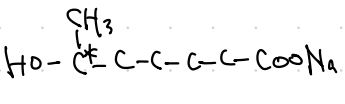
G : $C_7H_{12}O$　$U = 2$　 

非対称 → バイヤー・ビリガー酸化 → 主 H → NaOH → I の Na塩

I と NaOH

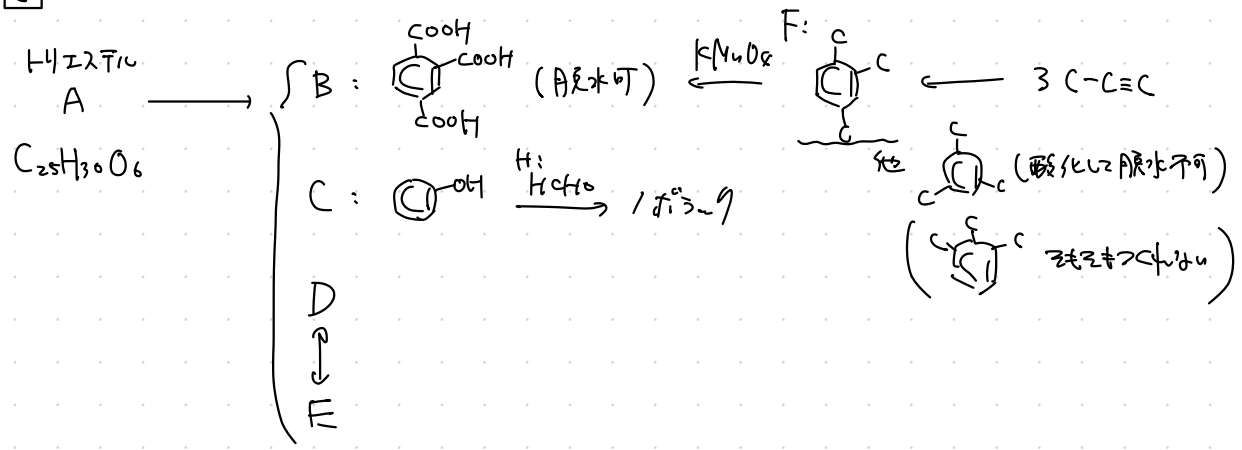
G : 

H : 

NaOH, △ → 

ア：e イ：c ウ：h エ：b

8



$$D\cdot E:\ \frac{C_{25}H_{30}O_6 + 3H_2O - C_9H_6O_6 - C_6H_6O}{2} = C_5H_{12}O$$

D：C−C−C−C*(H)−C

②

C−C−C=C−C
シス・トランス
C−C−C−C=C

E：C−C(C)−C−C
OH

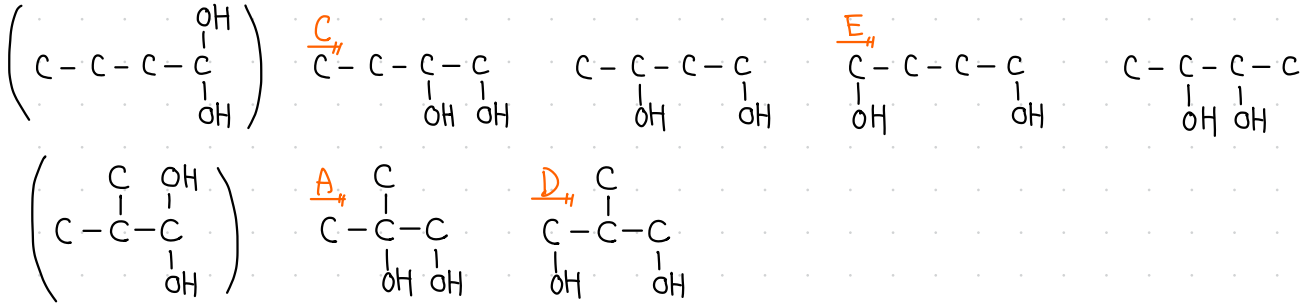
9

$(C_2H_5O)_n$ について、 $\underbrace{5n}_{\text{H数}} \leqq \underbrace{2\cdot 2n+2}_{\text{最大H数}}$ $\therefore n \leqq 2$ H数は偶数より、$n=2$.

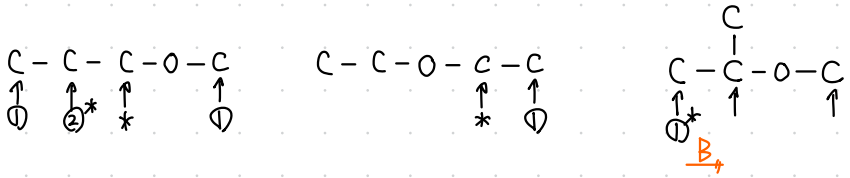
$\longrightarrow$ A〜E : $C_4H_{10}O_2$, $U=0$

$$\begin{cases} \text{A, C, D, E} : -OH \times 2 \\ \text{B} : -OH \times 1 + -O- \times 1 \end{cases}$$

〈A, C, D, E : −OH×2〉



〈B : −OH×1 + −O− ×1〉



1級アルコール + 3級

A $\xrightarrow[+14]{Ox}$ F

$-CH_2OH \longrightarrow -COOH$ ×1

A : $CH_3-C(CH_3)(OH)-CH_2-OH \xrightarrow{Ox}$ F : $CH_3-C(CH_3)(OH)-C(=O)-OH$

1か所酸化できないということは、3級アルコール構造あり

(−O−あり)

1級アルコール + B $\xrightarrow[+14]{Ox}$ G

$-CH_2OH \longrightarrow -COOH$ ×1

B : $CH_3-O-C^*H(CH_3)-CH_2-OH \xrightarrow{Ox}$ G : $CH_3-O-C^*H(CH_3)-C(=O)-OH$

1級ジオール + 2級 C $\xrightarrow[+12]{Ox}$ H

$-CH_2OH \longrightarrow -COOH$ ×1 + $-CH(OH)- \longrightarrow -CO-$ ×1

C : $CH_3-CH_2-C^*H(OH)-CH_2-OH$ (ヨウ素なし)

$\xrightarrow{Ox}$ H : $CH_3-CH_2-C(=O)-C(=O)-OH$

1級ジオール + 1級 D $\xrightarrow[+14\times 2]{Ox}$ ジカルボン酸 J

D $\xrightarrow{-2H}$ アルデヒド I*

$-CH_2OH \longrightarrow -COOH$ ×2

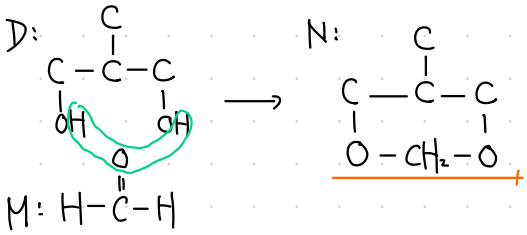
E : $C-C-C-C$ (OH, OH) $\xrightarrow[-2H]{Ox}$ K : $H-C(=O)-C-C-C-OH$

1級ジオール + 1級 E $\xrightarrow[+14\times 2]{Ox}$ ジカルボン酸 L

E $\xrightarrow{-2H}$ アルデヒド K

$-CH_2OH \longrightarrow -COOH$ ×2

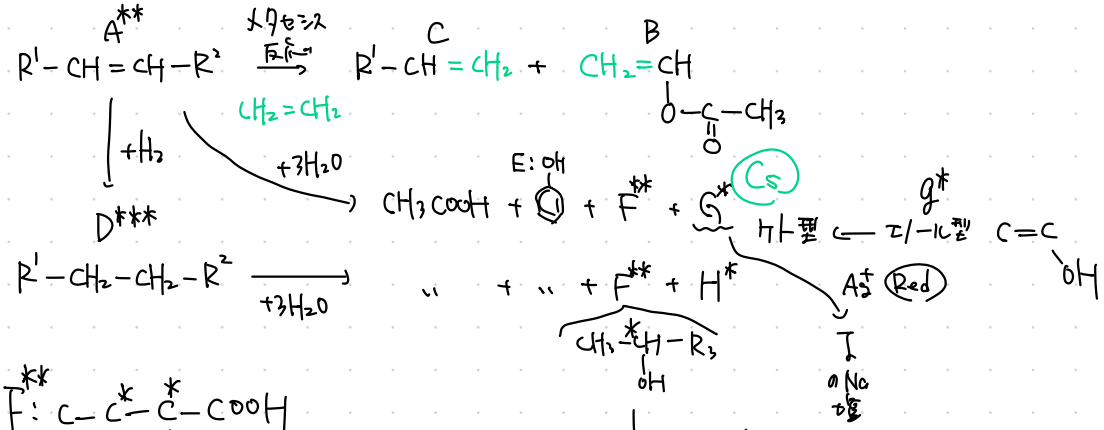
D : $C(-C(OH)-C-OH)(-OH)$ $\xrightarrow[-2H]{Ox}$ I* : $H-C(=O)-C^*(C)-C-OH$



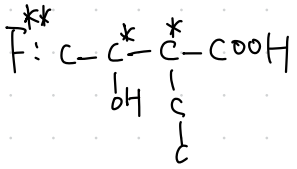
例10 449

10

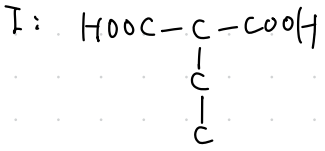
A**: $C_{19}H_{24}O_6$ ($U = 8 = -\overset{O}{\overset{\|}{C}}-O- \times 3 + \bigcirc \times 1 + C=C \times 1$)



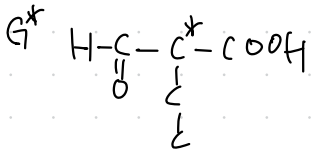
F**:



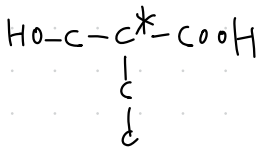
I:

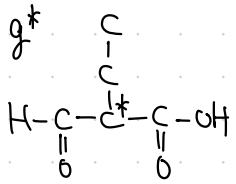


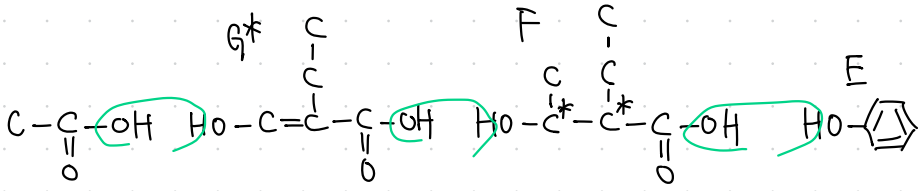
G*



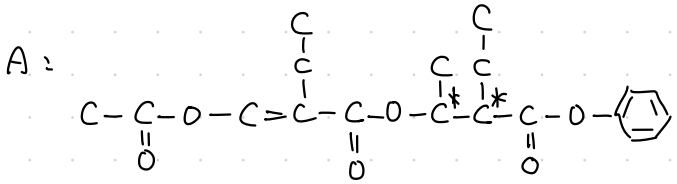
(H*)







A:



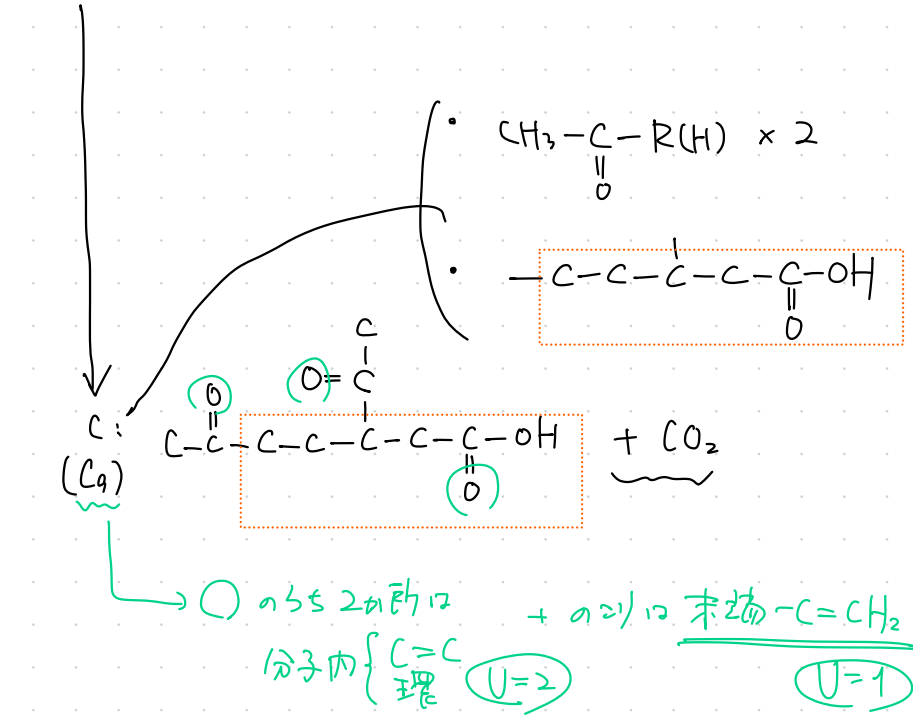
16

ア：$C_{10}H_{16}$

イ：＝重結合 2つ
または
≡重結合 1つ

ウ：3 4 3 3

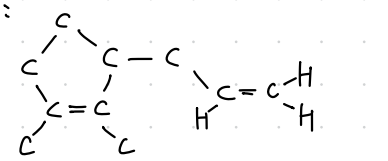
A：$C_{10}H_{16}$　$U=3$

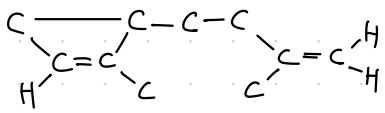


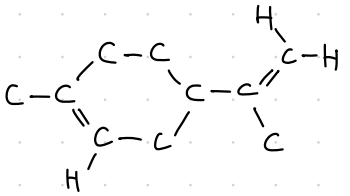
○のうち2か所は
分子内{C=C 環 $U=2$

＋の方は末端 $-C\equiv CH_2$
$U=1$

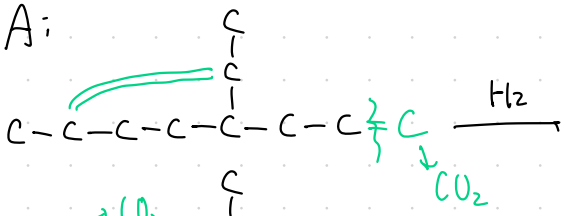
エ：



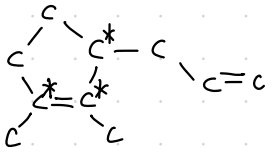


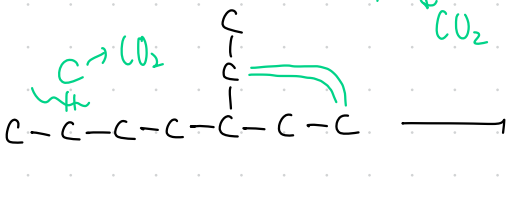


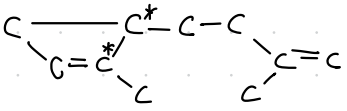
A：

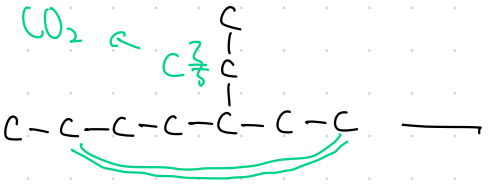


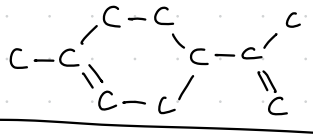
B











オ

